(iii) $25\times32\times625\times64 = (5\times5)\times(2\times2\times2\times2\times2)$
$\times(5\times5\times5\times5)\times(2\times2\times2\times2\times2\times2)$
$= 5^2\times2^5\times5^4\times2^6$
$= (5^2\times5^4)\times(2^5\times2^6)$ [grouping exponential numbers with the same base]
$= 5^{2+4}\times2^{5+6} = 5^6\times2^{11}$

3.3.2 Division of Numbers in Exponential form

Let us calculate the value of $2^5 \div 2^2$

$$\frac{2^5}{2^2} = \frac{2\times2\times2\times2\times2}{2\times2} = 2\times2\times2$$
$$= 2^3$$
$$= 2^{5-2}$$

We observe that the base of 2^5 and 2^2 is the same '2' and the difference of powers is 3. Now, let us consider negative integers as the base.

Consider $(-5)^3 \div (-5)^2$

$$\frac{(-5)^3}{(-5)^2} = \frac{(-5)\times(-5)\times(-5)}{(-5)\times(-5)}$$
$$= (-5)^1 = (-5)^{3-2}$$

We observe that the base of $(-5)^3$ and $(-5)^2$ is the same as (-5) and the difference of the power is 1.

Thus, we can observe that for any non-zero integer 'a' and for whole numbers 'm' and 'n', consider a^m and a^n, $m > n$.

That is, $a^m = a\times a\times a\times\ldots\times a$ (m times); $a^n = a\times a\times a\times\ldots\times a$ (n times)

$$\frac{a^m}{a^n} = \frac{a\times a\times a\ldots\times a(m\ times)}{a\times a\times a\ldots\times a(n times)} = a\times a\times a\ldots\times a(m-n\ times) = a^{m-n}$$

Therefore, $\dfrac{a^m}{a^n} = a^{m-n}$

This is called Quotient Rule of exponents.

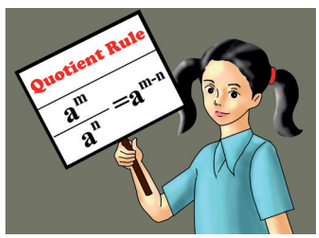


Fig. 3.4

Example 3.8 Simplify using quotient rule of exponents.

(i) $\dfrac{10^8}{10^6}$ (ii) $\dfrac{2^8\times3^5\times5^4}{3^3\times5^3\times2^4}$ (iii) $\dfrac{6^4}{6^0}$

Solution

(i) $\dfrac{10^8}{10^6} = 10^{8-6} = 10^2$

Note

Can you find the value of $a^2\times a^0$?

By Product Rule,

$$a^2\times a^0 = a^{2+0}$$
$$a^2\times a^0 = a^2$$
$$a^0 = \frac{a^2}{a^2} = 1$$

[dividing by a^2 on both sides]

Therefore, $a^0 = 1$, $a \neq 0$.

Think

What is Half of 2^{10}? Ragu claims the answer is 2^5. Is he correct? Discuss.

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